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Antidepressant-like effect of peony glycosides in mice

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ABSTRACT

Aim of the study: The root part of *Paeonia lactiflora* Pall. (Ranunculaceae), known as peony, is often used in Chinese herbal formulae for the treatment of depression-like disorders. Previous studies in our laboratory have shown that an ethanol extract of peony produced antidepressive effects in mouse models of depression. It is well known that peony contains glycosides such as paeoniflorin and albiflorin, yet it remains unclear whether the total glycosides of peony (TGP) are effective. The present study aims to evaluate the antidepressant-like effects of TGP.
Materials and methods: The antidepressant-like effects of TGP was determined by using animal models of depression including forced swim and tail suspension tests. The acting mechanism was explored by determining the effect of TGP on the activities of monoamine oxidases.
Results: Intragastric administration of TGP at 80 and 160 mg/kg for seven days caused a significant reduction of immobility time in both forced swim and tail suspension tests, yet TGP did not stimulate locomotor activity in the open-field test. In addition, TGP treatment antagonized reserpine-induced ptosis and inhibited the activities of monoamine oxidases in mouse cerebrum.
Conclusion: These results suggest that the antidepressive effects of TGP are mediated, at least in part, by the inhibition of monoamine oxidases.

1. Introduction

World Health Organization predicts that depression will become the second leading cause of disability in both developed and developing countries by the year 2020 (Kim et al., 2007). At present, there are several types of classical antidepressants in clinical practice, including tricyclic antidepressants (TCA), monoamine oxidase inhibitor (MAOI), selective serotonin reuptake inhibitor (SSRI), noradrenergic reuptake inhibitor (NARI) and serotonin and noradrenaline reuptake inhibitor (SNRI) (Bouvier et al., 2003; Shen and Liang, 2007). Most of these drugs, however, have undesirable side effects. Thus, there is an unmet need for safe, better tolerated and powerful antidepressants.

Traditional Chinese medicine has recorded many herbal formulae with psychotropic potential. In recent year, herbal preparations have been evaluated in animal models of depression (Zhang, 2004). The root part of *Paeonia lactiflora* Pall. (Family: Ranunculaceae), commonly known as peony, is one of the most commonly used medicinal herbs in China, Korea and Japan. It is a component herb of many traditional formulae (for example, Sini-San and Xiaoyao-San) for the treatment of depression-like disorders (Zhang et al., 1998; Xie and Wang, 2005). Previous studies in our laboratory have demonstrated that an ethanol extract of peony displayed antidepressive effects in mouse models of depression (Mao et al., 2008). Glycosides such as paeoniflorin and albiflorin are known to be the biologically active ingredients of peony, and its total glycosides have been shown to possess anti-inflammatory, antioxidant, immunomodulatory, antithrombotic and neuroprotective properties (Zhou et al., 1994; Liu et al., 2001; Dong et al., 2003; Yang et al., 2006; Xu et al., 2007). However, it remains unknown whether the total glycosides of peony (TGP) are the active principles for antidepressive effect. In this study, we aim to evaluate the antidepressant-like effects of TGP by using the forced swim and tail suspension tests. The underlying mechanism of antidepression is explored by testing the antagonizing effect against reserpine-induced ptosis and measuring the activity of monoamine oxidase (MAO) in mouse cerebrum.

2. Materials and methods

2.1. Drugs and chemical reagents

The total glycosides of peony (light yellow brown powder) were supplied by Ningbo Liwah Pharmaceutical Co., Ltd. (Zhejiang, China). The method of preparation was described previously (Zhao et al., 1997). The product was analyzed and shown to

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contain 30% (w/w) of paeoniflorin and 10% (w/w) of albiflorin as determined by high-performance liquid chromatography. The major chemical ingredients of TGP were consistent with previous studies (Wang et al., 2005; Xu et al., 2007). A voucher sample (TGP071024) has been deposited in the School of Chinese Medicine for future reference. Chlorimipramine and moclobemide were purchased from Beijing Novartis Pharmaceutical Co., Ltd. (Beijing, China) and Shanghai Xinyi Pharmaceutical Co., Ltd. (Shanghai, China) respectively; reserpine solution was produced by Guangdong Bangming Pharmaceutical Co., Ltd. (Guangdong, China); 5-hydroxytrylamine (5-HT) and β-phenylethylamine (β-PEA) were obtained from Sigma–Aldrich (St. Louis, MO, USA). All other reagents and solvents used in the study were of analytical grade.

2.2. Animals

Male ICR mice weighing 20–25 g were obtained from the Laboratory Animal Services Center, The Chinese University of Hong Kong, Hong Kong. The animals were maintained on a 12-h light/dark cycle under regulated temperature (22 ± 2 °C) and humidity (50 ± 10%) and fed with standard diet and water ad libitum. They were allowed to acclimate three days before use. All animal care procedures were conducted in accordance with the NIH Guidelines for the Care and Use of Laboratory Animals.

2.3. Drug administration

The animals were randomly assigned into groups of fifty individuals. Distilled water was given to animals in group 1 (control). Animals in groups 2, 3 and 4 received intragastric doses of peony glycosides at 40, 80 and 160 mg/kg, respectively. Animals in group 5 were administered with reference compounds (chlorimipramine or moclobemide, 20 mg/kg). The drugs were given daily between 9:30 and 10:30 a.m. for seven days. Sixty minutes after the last dose, ten mice in each group were used to perform the behavioral tests, and the other ten mice were sacrificed for the determination of monoamine oxidase activity.

2.4. Forced swim test

The test was carried out according to the method described by Porsolt et al. (1977) with modifications. Briefly, mice were forced to swim in a transparent glass vessel (25 cm high, 14 cm in diameter) filled with 10 cm of water at 24–26 °C. The total duration of immobility (seconds) was measured during the last 4 min of a single 6-min test session. Mice were considered immobile when they made no attempts to escape except the movements necessary to keep their heads above the water.

2.5. Tail suspension test

Tail suspension test was performed based on the method of Steru et al. (1985). Briefly, mice were suspended 5 cm above the floor by means of an adhesive tape, placed approximately 1 cm from the tip of the tail. The total duration of immobility (s) was quantified during a test period of 6 min. Mice were considered immobile when they were completely motionless.

2.6. Open-field test

The ambulatory behavior was assessed in an open-field test as described previously (Herrera-Ruiz et al., 2006). The open-field apparatus consisted of a square wooden arena (30 cm × 30 cm × 15 cm) with black surface covering the inside walls. The floor of the wooden arena was divided equally into twenty-five squares marked by black lines. In the test, each mouse was placed individually into the center of the arena and allowed to explore freely. The number of squares crossed by the mouse and the number of rearings on the hind paws were recorded during a test period of 3 min. This apparatus was cleaned with a detergent and dried after occupancy by each mouse.

2.7. Reversal of reserpine-induced ptosis in mice

The test was performed according to the method described by Bourin et al. (1983) with modifications. Reserpine (2.5 mg/kg) was given intraperitoneally to the animals and ptosis was evaluated 120 min after reserpine treatment. Animals were placed on a shelf (20 cm above the tabletop) and the degree of ptosis was rated according to the following rating scale: 0, eyes open; 1, eyes one-quarter closed; 2, eyes half closed; 3, eyes three-quarters closed; and 4, eyes completely closed (Sánchez-Mateo et al., 2007).

2.8. Monoamine oxidase (MAO) assay

Cerebral tissue samples were collected from sacrificed animals, rapidly frozen and stored at −80 °C until analyzed. Mouse cerebral MAO activity was measured following the procedures described previously (Yu et al., 2002; Zhou et al., 2006). Briefly, the cerebral tissue was suspended in 10 volume of cold sodium phosphate buffer (200 mM, pH 7.4) containing 320 mM sucrose and homogenized at 4 °C for 30 s by using a motor-driven pellet pestle. The mixture was centrifuged at 600 × *g* for 10 min at 4 °C to remove nucleus and cell debris. The mitochondrial fraction was obtained by further centrifugation at 15,000 × *g* for 30 min at 4 °C and resuspended in buffer. Protein concentration was determined by the Lowry method (Lowry et al., 1951) using bovine serum albumin as the standard. The protein concentration of the mitochondrial fraction was diluted to 1 mg/ml. The MAO assay mixture contained 400 μl of mitochondrial protein in the phosphate buffer. 5-Hydroxytrylamine (5-HT, 4 mM) or β-phenylethylamine (β-PEA, 2 mM) was added as specific substrate for monoamine oxidase A (MAO-A) and monoamine oxidase B (MAO-B), respectively. The final volume of the reaction mixture was 3 ml. The reaction mixture was incubated at 37 °C for 60 min, followed by the addition of HCl (600 μl, 1 M). The reaction products were extracted by 4 ml of butylacetate or cyclohexane for the assay of MAO-A and MAO-B, respectively. The organic phase was collected and the absorbance was measured at 280 nm (for MAO-A) and 242 nm (for MAO-B), respectively. Blank samples were prepared by adding 600 μl of 1 M HCl prior to reaction, and worked up subsequently in the same manner.

2.9. Statistical analysis

Data were expressed as mean ± S.E.M. Multiple group comparisons were performed using one-way analysis of variance (ANOVA) followed by Dunnett's test in order to detect inter-group differences. Difference was considered statistically significant when the $p < 0.05$.

3. Results

The antidepressive effects of the total glycosides of peony (TGP) were evaluated in two models of depression in mice. Treating the animals with TGP for seven days reduced the duration of immobility in the forced swim test in a dose-dependent manner, although the difference from the control at the lowest dose (40 mg/kg) was not statistically significant. When compared with the control group,

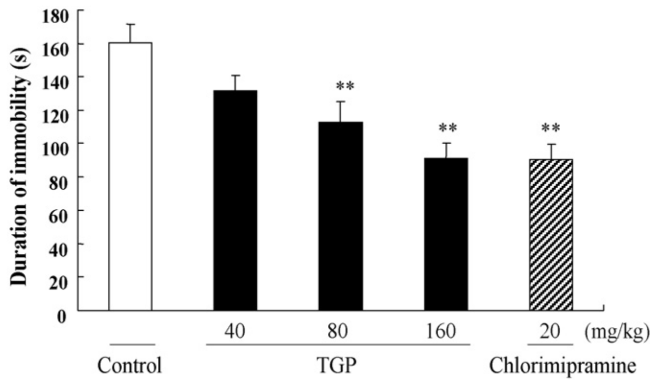

Fig. 1. Effect of total glycosides of peony (TGP) and chlorimipramine on immobility time of mice in forced swim test. Values given are the mean ± S.E.M. ($n = 10$). $^{**}p < 0.01$ as compared with control.

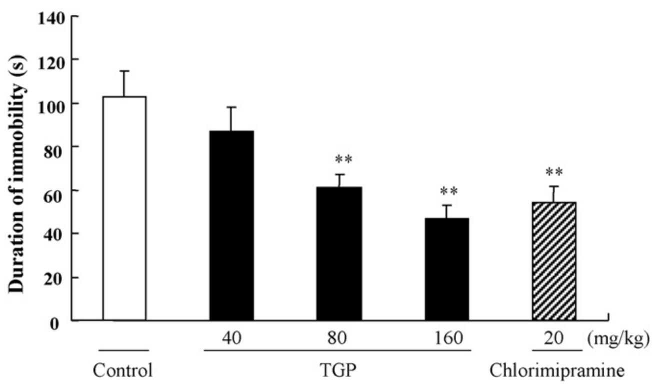

Fig. 2. Effect of total glycosides of peony (TGP) and chlorimipramine on immobility time of mice in tail suspension test. Values given are the mean ± S.E.M. ($n = 10$). $^{**}p < 0.01$ as compared with control.

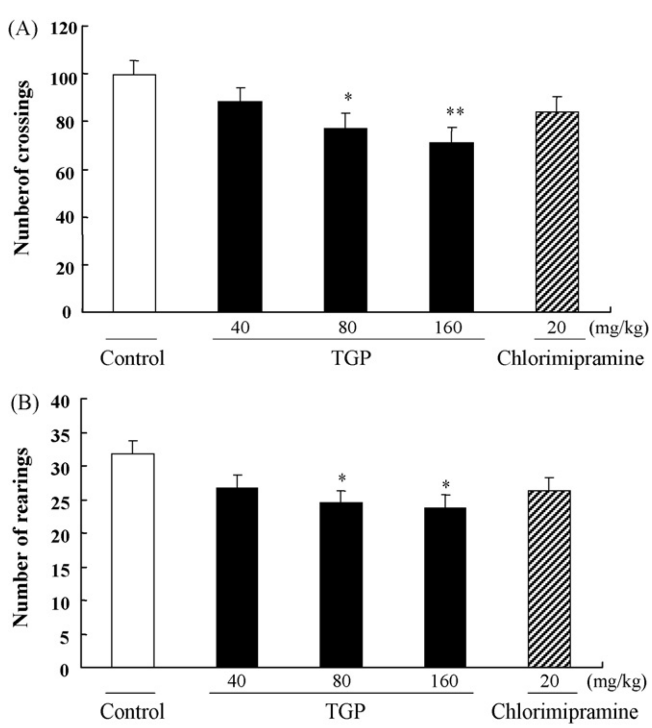

Fig. 3. Effect of total glycosides of peony (TGP) and chlorimipramine on the number of crossings (A) and rearings (B) in the open-field test. Values given are the mean ± S.E.M. ($n = 10$). $^{*}p < 0.05$, $^{**}p < 0.01$ as compared with control.

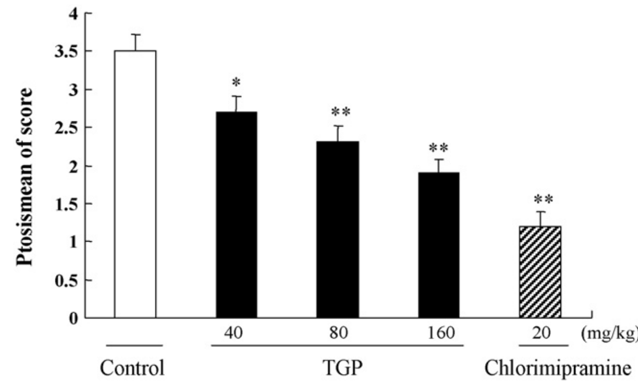

Fig. 4. Effect of total glycosides of peony (TGP) and chlorimipramine on antagonism of reserpine-induced ptosis in mice. Values given are the mean ± S.E.M. ($n = 10$). $^{*}p < 0.05$, $^{**}p < 0.01$ as compared with control.

the duration of immobility was decreased by 17.7%, 29.7% and 43.2% for doses of 40, 80 and 160 mg/kg, respectively (Fig. 1). The same treatment regimen with TGP at doses of 80 and 160 mg/kg also significantly decreased the immobility time in the tail suspension test. The duration of immobility was reduced respectively by 40.7% and 54.6% when compared with the control (Fig. 2). The classical antidepressant chlorimipramine, used as a positive control, caused a significant reduction in the immobility time in both models. Treating the mice with chlorimipramine at daily dose of 20 mg/kg for seven days caused a reduction of immobility time by 43.6% and 47.4% in the forced swim test and tail suspension test, respectively (Figs. 1 and 2).

The effect of TGP on ambulatory activity was evaluated by open-field behavior test. The results showed that TGP or chlorimipramine did not increase the number of crossings and rearings in the open-field test (Fig. 3). Instead, TGP treatment at doses of 80 and 160 mg/kg caused a slight reduction of locomotor activity.

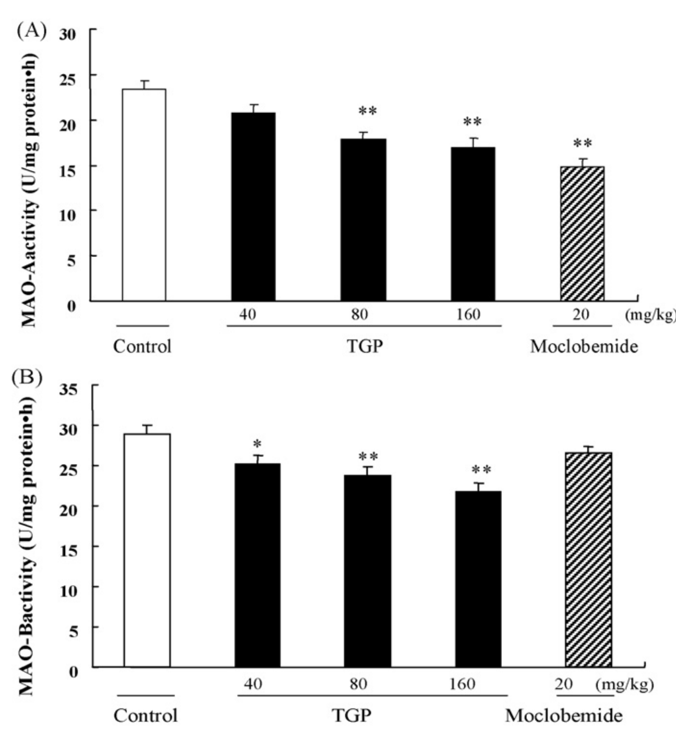

Fig. 5. Effect of total glycosides of peony (TGP) and moclobemide on enzyme activities of MAO-A (A) and MAO-B (B) in mouse cerebrum. Values given are the mean ± S.E.M. ($n = 10$). $^{*}p < 0.05$, $^{**}p < 0.01$ as compared with control.

The effect of TGP on reserpine-induced ptosis in mice was shown in Fig. 4. Treating the mice with TGP at daily doses of 40, 80 or 160 mg/kg for seven days significantly antagonized the reserpine-induced ptosis in a dose-dependent manner. The same treatment protocol using chlorimipramine at 20 mg/kg also significantly antagonized ptosis induced by reserpine.

The effects of TGP and moclobemide on the enzyme activities of monoamine oxidases (MAO-A and MAO-B) in mouse cerebrum were shown in Fig. 5. Comparing with the control group, TGP at 80 and 160 mg/kg significantly inhibited the activity of MAO-A. It also showed a dose-dependent inhibition of MAO-B. The same treatment regimen with moclobemide (a reversible inhibitor of MAO-A) at 20 mg/kg inhibited the activity of MAO but not MAO-B.

4. Discussion

Forced swim test and tail suspension test are the widely used animal models of depression for the screening of antidepressive activity (Porsolt et al., 1977; Steru et al., 1985). In these tests, animals are under stress from which they cannot escape. After an initial period of struggling, they would become immobile, resembling a state of despair and mental depression (Porsolt et al., 1977; Steru et al., 1985). In the present study, the antidepressive effects of TGP were evaluated by using these animal models. After being treated with the herbal preparation at 80 and 160 mg/kg for seven days, the mice showed a significant reduction of immobility time in both forced swim (Fig. 1) and tail suspension tests (Fig. 2).

In these behavioral tests, false-positive results can be obtained for agents that stimulate locomotor activity (Bourin et al., 2001). Therefore, the effect of TGP on locomotion was evaluated by the open-field test. The results showed that TGP or chlorimipramine treatment did not increase the number of crossings and rearings (Fig. 3). Instead, at high doses of TGP (80 and 160 mg/kg), the number of crossings and rearings was slightly decreased after treatment. The finding suggested that the reduction of immobility time elicited by TGP treatment in the forced swim test and tail suspension test was unlikely due to a psychomotor-stimulant effect, but rather an antidepressant-like effect of TGP.

We then tried to explore the antidepressive mechanism of TGP by testing the reserpine-induced ptosis. Reserpine irreversibly inhibits the vesicular uptake of monoamines neurotransmitters (including noradrenaline, dopamine and 5-hydroxytrytamine), which are subsequently metabolized by monoamine oxidases. As a consequence, ptosis is observed in animals suffering from a depletion of monoamines (Bourin et al., 1983; Dhingra and Sharma, 2006). The symptom can be reversed by major classes of antidepressant drugs. The results obtained in the present study indicated that pretreating mice with TGP dose-dependently antagonized the ptosis induced by reserpine. Such an observation suggested that the antidepressant-like effect of TGP may be caused by the preservation of monoamine neurotransmitters.

Preservation of monoamine neurotransmitters can be achieved by the inhibition of monoamine oxidases (MAO). It was demonstrated that high doses of TGP (80 and 160 mg/kg) significantly inhibited the activity of MAO-A in mouse cerebrum, whereas the activity of MAO-B was inhibited dose-dependently at all three dose levels. These findings suggested that the antidepressive effect of TGP may be mediated by the inhibition of MAO.

In conclusion, an antidepressant-like effect of TGP was demonstrated in animal models of depression (forced swim test and tail suspension test). The effect is mediated, at least in part, by an inhibition of monoamine oxidases. Further study is warranted to reveal the antidepressive mechanism of TGP.

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